Assignment - 4

⇒ Aim

Write a Java program to showcase the use of File I/O & Exception Handling in Java.

⇒ Objectives

- To study I/O stream classes
- ~~Explain~~ Understand the different types of
- Exception Handling using various Exception classes.

⇒ Theory

1) Java I/O Streams

Java provides I/O streams to read & write data

- Byte Streams -
  Handle raw binary data.
- Character Streams -
  Handle Unicode characters.

2) Exception Handling Mechanism

Exceptions are events that disrupt the normal flow of a program.

Java uses five key keywords: try, catch, finally, throw, throws.

- Checked Exceptions -
  Verified at compile-time.
- Unchecked Exceptions -
  Verified at runtime.

⇒ Algorithm

1) Define Customer Class -
Create a class to store cid, cname, & amount.

2) Define Custom Exceptions
Create classes for specific banking violations.

3) Input & Validation -
- Valid that cid is between 1 & 20
- Ensure the initial deposit is at least Rs 1000
- Ensure all entered amounts are positive.

4) Transaction Logic -
- During withdrawal, check if withdrawal amount > total amount, if so throw an exception.

5) File Persistance -
Use ObjectOutputStream to save validated customer records to a file.

6) Menu Loop -
Provide options to Create Account, Deposit, withdraw, & Exit.

⇒ Conclusion
Successfully implemented a banking system that utilizes Exception Handling to ensure data integrity & File I/O to persist customer records.

⟹ FAQs

1) Difference between Error & Exception?

- Error -
Indicates serious problems that a reasonable application should not try to catch.
- Exception -
Indicates conditions that an application might want to catch.

2) Difference between throw & throws?

- Throw -
Used to explicitly throw an exception instance.
- Throws -
Used in a method signature to declare that the method may throw certain exceptions

3) Keywords in Exception Handling?

- Try, catch, finally, throw & throws.

4) What is the difference between throws & throw keywords in Java?

→ • Throw -
Used to explicitly throw a single exception from within a method or block
- Throws -
Used in a mixed signature to declare that the method might throw one or more exceptions.

5) List down various byte & character stream classes in Java?

→ • Byte Streams →
Fileinputstream, Fileoutputstream, Bufferinputstream, BufferedOutputstream.

• Character Streams —
FileReader, FileWriter, BufferedReader, BufferedWriter.